

Liberal Arts Colleges: Gap Fixes and Supplementary Research

Date: 2026-05-23 **Purpose:** Address gaps identified in `research-notes-liberal-arts-colleges.md`

Method: Source-verified synthesis. Each section maps to a specific gap in the original document.

Gap Fix 1: The Middle-Income Squeeze — The Decision Families Actually Face

1.1 The Problem

The original document shows the Williams subsidy (\$54,400 for full-pay students) and Pell data for low-income students but is silent on families earning \$100K–\$250K — the group most likely to be reading this research.

1.2 The Bloomberg Analysis (April 2025)

Bloomberg analyzed financial aid data from 50 selective colleges using MyinTuition calculator estimates:

“The squeeze starts at about \$150,000 of pre-tax household income, when families are expected to fork over roughly 20%, or about \$30,000 annually, for tuition. At \$270,000 of income families are expected to pay \$61,000 a year. Most schools cut off financial aid at about \$400,000 of income.”

“At Williams College, a student with \$300,000 of family income would be asked to pay from \$43,000 to \$73,000 a year toward the roughly \$92,000 sticker price.”

“Nearly half of first-year students at the top schools in the US pay full price, meaning the students come from households with annual incomes at roughly \$400,000 or above.”

A key and counterintuitive finding:

“Families making \$75,000 to \$150,000 can actually pay less at private colleges, which can offset higher tuition with more generous aid.”

“At the top of that income range [\$75K–\$150K], families are expected to pay roughly \$35,000 on average at 15 public schools — roughly 20% more than the expected contribution at private universities.”

This means the *real* middle-income squeeze hits families earning \$150K–\$400K, not those below \$150K. For families below \$150K, elite LACs with large endowments can actually be *cheaper* than state flagships.

Source: Bloomberg, “Top US College Costs Squeeze Middle Class, Even Parents Making \$300,000,” Ann Choi, Francesca Maglione, Paulina Cachero, Raedah Wahid, April 25, 2025. <https://www.bloomberg.com/graphics/2025-college-cost-middle-class-squeeze/>

1.3 Bowdoin’s Published Net Price by Income (Institutional Data)

Bowdoin is one of the few top LACs to publish a granular net price table for aided students:

Total Family Income	Median Net Price
\$0–\$30,000	\$4,700
\$30,001–\$48,000	\$4,700
\$48,001–\$75,000	\$9,100
\$75,001–\$110,000	\$17,500
\$110,001–\$150,000	\$23,400
\$150,001–\$200,000	\$36,834
\$200,001 or more	\$47,250

Note: These are median figures for students receiving Bowdoin need-based aid. The full sticker price for 2024–25 was ~\$84,000.

Source: Bowdoin College, “Net Price by Income.” <https://www.bowdoin.edu/student-aid/estimate-aid/net-price-by-income.html>

1.4 The Wellesley/Mildly Efficient Case Study (2024)

Seth Neumuller, Associate Professor of Economics at Wellesley College, analyzed a decade of financial data at an unnamed top-tier LAC (widely inferred to be Wellesley):

- **Sticker price (2024–25):** ~\$84,000
- **Average family paid:** ~\$48,000 (a 40% markdown)
- **55% of students** received need-based aid
- **Average grant for aided students:** ~\$67,000
- **Aided families’ average bill:** ~\$17,000 (an 80% discount)
- **Full-pay families (45%):** ~\$84,000
- **Per-student operating expenses:** >\$115,000
- **Even full-pay families received an implicit subsidy** of ~27% off the true cost
- **Aided families’ bills fell ~1.4% per year** even as sticker prices rose ~4% per year

The gap between aided and full-pay families grew from ~\$40,000 in 2015 to ~\$67,000 in 2024, a 68% increase — what Neumuller calls “increasing price dispersion.”

Source: Seth Neumuller, “The Real Cost of a Top-tier Liberal Arts Education,” *Mildly Efficient* (Substack), September 2024. <https://www.mildlyefficient.com/p/the-real-cost-of-a-top-tier-liberal>

1.5 Institutional Responses to the Squeeze

Several institutions expanded aid for middle-income families in 2024–2025:

- **Harvard, MIT, UPenn:** Expanded free tuition to families earning up to \$200,000.

- **Johns Hopkins:** Michael Bloomberg’s \$1.8 billion donation enabled covering tuition for the majority of students, and eliminating student loans from all aid packages.
- **Colby College:** Capped tuition, room, and board at \$10,000/year for families earning up to \$100,000, and \$15,000 for \$100K–\$150K.

Sources: Bloomberg (December 2024); NPR, “Some colleges are targeting financial aid to middle-class families,” October 22, 2024; JHU Newsletter, November 2025.

1.6 What This Means for the LAC Assessment

The original document’s financial section (Section 3) is accurate about endowment strength but misleading by omission. The correct framing:

- **Families earning <\$75K:** Elite LACs are often free or near-free. They can be cheaper than public flagships.
- **Families earning \$75K–\$150K:** Elite LACs can be comparable to or cheaper than out-of-state flagships, though sometimes slightly more expensive than in-state options.
- **Families earning \$150K–\$300K:** This is the genuine squeeze zone. These families pay \$30K–\$73K/year at top LACs — a meaningful burden but well below sticker price.
- **Families earning >\$400K:** Pay full sticker price (~\$85K–\$92K). They represent ~45% of students at top LACs.

The decision-relevant question is not “is Williams affordable?” but “for a family earning \$X, is the net price difference between Williams and Michigan with merit aid justified by the outcomes difference?” — a question the original document does not address.

Gap Fix 2: Mental Health — What Gallup Doesn’t Capture

2.1 The Gap

Section 12 of the original document discusses alumni wellbeing using Gallup data but says nothing about mental health *during* the college years. This is a significant omission for a document evaluating the student experience.

2.2 Healthy Minds Study: National Data

The Healthy Minds Study (HMS) is the largest mental health survey in U.S. higher education, conducted by the Healthy Minds Network (based at University of Michigan, UCLA, and Wayne State). It has been administered on 800+ campuses since its inception.

2024–2025 Healthy Minds Study (84,735 students across 135 colleges and universities):

- **32% of students** reported moderate-to-severe anxiety
- **22% reported severe depression** (down from 23% in 2022)
- **18% reported severe depressive symptoms** (down from 23% in 2022)
- **11% reported suicidal ideation** (down from 15% in 2022)
- **37% of students** received therapy or counseling in the past year
- **68% reported** mental health difficulties impacting academic performance at least one day that month

A Johns Hopkins analysis of HMS data (2007–2022, 560,000+ students) found suicidal ideation up 154%, restlessness up 80%, and trouble concentrating up 77% over the study period.

Sources: Healthy Minds Study 2024–2025 Data Report, UCLA Center for Health Policy Research, September 2025; University of Michigan School of Public Health, “College Student Mental Health Shows Third Year of Improvement,” September 2025; The Hill, “Colleges struggle to keep up with growing mental health problems,” March 2026; Inside Higher Ed, “College Student Mental Health Remains Poor,” September 2025.

2.3 Trends: Improving, But Still Elevated

The 2024–25 data shows improvement for the third consecutive year since the COVID-19 peak. But rates remain far above pre-2020 baselines:

“One-third of university students were reporting clinical anxiety and two-fifths were reporting clinical levels of depression... adult population had prevalence rates of 6 to 7 percent.” — Leslie Rith-Najarian, UCLA

“About 1 in 4 students who experience mental health challenges drop out of school, with those who experience depression being twice as likely to leave college without completing a degree.” — U.S. News/Brian Bridges, October 2024

2.4 Relevance to LACs Specifically

The HMS does not publish institution-type breakdowns in its public reports, making LAC-specific comparisons difficult. However, two dynamics are relevant:

Arguments that LACs may do better: - Small, tight-knit communities with higher faculty-student interaction - Higher counselor-to-student ratios at well-resourced institutions - Near-zero dropout rates suggest distressed students are retained and supported

Arguments that LACs may do worse: - The “pressure cooker” dynamic of a small, hyper-competitive, residential environment with no anonymity - Jennifer Breheny Wallace’s *Never Enough* (2023) documents the achievement culture that drives students to elite institutions, arriving “prestressed” from competitive high school environments - Limited ability to “escape” the social fishbowl of a 2,000-student campus - Strong selection for perfectionism and high-achievement orientation

Critical assessment: Without LAC-specific HMS data, claims in either direction are speculative. The Gallup finding that small-school graduates were “slightly less likely to be engaged at work” (Section 12.1 in the original) may point to something — but the connection between college-era mental health and post-graduation engagement is not established.

Sources: Jennifer Breheny Wallace, *Never Enough: When Achievement Culture Becomes Toxic — and What We Can Do About It*, Portfolio/Penguin, 2023. Reviewed in Inside Higher Ed, January 2024.

Gap Fix 3: Athletics as Admissions Architecture

3.1 The Gap

The original document does not mention athletics at all. At top LACs, this is a defining compositional feature.

3.2 Scale of Athletic Participation

Williams College provides the clearest data:

“More than 60% of Williams’ students compete on at least one varsity or club team.”

“Approximately 31% of students participate in varsity sports in a given year.”

Williams fields **32 varsity intercollegiate teams** (16 men’s, 16 women’s), plus 16 club teams and intramurals.

Source: Williams College Fast Facts, Office of Communications, Fall 2025. <https://communications.williams.edu/media-relations/fast-facts/>; Williams Athletics, “About.” <https://athletics.williams.edu/about/>

3.3 The NESCAC Banding System

A 2014 investigation by the *Bowdoin Orient* revealed the NESCAC’s systematic athletic admissions process, first disclosed in a 2005 *New York Times* article:

- Each NESCAC institution is allowed a maximum of **14 recruited athlete slots** for having a football team, with an additional **2 per remaining varsity sport**
- At Amherst, this totaled **66 recruits** per class — students for whom “the athletic input controls the decision”
- At Bowdoin (31 varsity sports), the estimated total was approximately **75–77 recruited spots**, or ~15% of the incoming class
- Recruits are divided into “bands”: **A-band** (within the student body average — not counted as athletic admits), **B-band** (slightly below average), and **C-band** (lower but above a minimum threshold)
- Each school has “a very limited number of C bands” it can use

Former Amherst Dean of Admissions Thomas Parker:

“In those 66 cases, the athletic input controls the decision. You have to say that in that group of 66 students, preference was given to them in the process, no question about it.”

“There’s a point at all the NESCAC schools when you can’t make any more academic distinctions because everybody is so good.”

The mean SAT score for athletes filling NESCAC slots was **60–75 points below** the mean for the freshman class at Amherst (as of 2005).

Source: *Bowdoin Orient*, “Banded together: recruited athletes with sub-average academics get preferential admissions treatment,” 2014. <https://bowdoinorient.com/bonus/article/9151/>; *New York Times*, December 2005 (cited in *Orient* investigation).

3.4 Competitive Dominance

Williams has been a perennial top-3 finisher in the NCAA Division III LEARFIELD Directors’ Cup, which ranks the best overall athletics programs. In 2024–25, the standings showed:

- All 11 NESCAC institutions finished in the top 100 nationally — the **25th consecutive year** this has occurred
- Tufts 3rd, Middlebury 5th, Amherst 7th, Williams 11th, Wesleyan 19th

Source: NESCAC, “All 11 NESCAC Institutions Ranked in 2024-25 LEARFIELD Directors’ Cup Final Standings,” June 2025. <https://nescac.com/news/2025/6/10/>

3.5 Why This Matters for the LAC Assessment

The athletic composition has several implications:

1. **Acceptance rate distortion:** When 15% of each class enters through athletic recruitment with preferential admissions, the “effective” acceptance rate for unhooked academic applicants is substantially lower than published figures suggest. One admissions analysis estimated Williams’ effective RD acceptance rate for unhooked applicants at **4–6%**.
2. **Socioeconomic composition:** D3 recruited athletes at NESCAC schools skew white and upper-middle-class. Many sports (lacrosse, squash, crew, field hockey, skiing) have high participation barriers. This partly explains why white enrollment has been relatively stable post-SFFA while Black enrollment has dropped (Section 3E of the original).
3. **Campus culture:** When 31–60% of students are varsity or club athletes, athletics defines the social fabric in ways that “liberal arts” branding may not convey. This is neither good nor bad, but it is a fact that prospective students should understand.
4. **The identity tension:** A school where 60% of students are athletes, CS is a top major, and 26.4% of graduates go into finance has a lived experience substantially different from the “small college of the mind” that the liberal arts brand evokes.

Source: Oriel Admissions, “Williams Acceptance Rate: Class of 2030 Trend,” May 2026.

Gap Fix 4: Test-Optional and Its Distortion of Selectivity Metrics

4.1 The Gap

Section 2 of the original presents acceptance rates and SAT mid-50% ranges as straightforward. But most top LACs went test-optional during COVID. This creates two systematic distortions.

4.2 Application Volume Inflation

Research from the University of Georgia (cited by the Hechinger Report):

“By freeing prospective students from having to provide SAT and ACT scores, [test-optional schools] tend to attract more applicants, many of whom may have scored poorly on the tests. For the colleges, more applicants mean more students they can reject, which lowers their acceptance rate and raises the institution’s perceived selectivity.”

Schools saw approximately **220 more applications on average** after adopting a test-optional policy.

At the elite LAC level, the effect is larger. Amherst received over 2,100 additional applications in a single year after maintaining test-optional. Bowdoin’s applications surged 20% in one cycle.

Source: Hechinger Report, “The real reason that colleges go ‘test-optional,’” 2019 (citing University of Georgia research); Ivy Coach, “Selective Colleges Reinstate Testing, See Drop in Applications,” April 2025.

4.3 SAT Score Inflation

“After going test-optional, these schools saw their mean SAT scores rise, on average, by 26 points.”

This happens because only high-scoring students submit scores. The published mid-50% ranges in Section 2 of the original document reflect only those who chose to submit — a self-selected, upward-biased sample. U.S. News will not count test scores if fewer than 75% of applicants submit, but many schools hover just above or below this threshold.

Source: University of Georgia study, cited in Hechinger Report (2019); *American Sociological Review* (Hsu & Sharkey, 2025).

4.4 Peer-Reviewed Evidence

A 2025 study published in *American Sociological Review* examined 180 selective liberal arts colleges:

“Adopting test-optional policies did not increase racial or socioeconomic diversity. After adopting test-optional policies, the liberal arts colleges in the study received an increased number of applications thus decreasing their acceptance rates.”

“Adopting a test-optional policy may boost the academic profile of an institution, as only reported test scores are used to calculate averages. However, this may have the unintended consequence of dissuading historically underserved students from applying due to perceived selectivity.”

An EdWorkingPapers study found:

“At highly selective colleges, test-optional policies were associated with increased applications, but we did not find consistent evidence of concurrent increases in enrollment.”

Sources: Hsu G, Sharkey A. “Same Policy, No Standardized Outcome: How Admissions Values and Institutional Priorities Shape the Effect of Test-Optional Policies on Campus Diversity.” *American Sociological Review*, 2025; Belasco AS, Rosinger KO, Hearn JC (2015), cited in Hsu & Sharkey; EdWorkingPapers.com, “Exploring the Relationship Between Test-Optional,” 2024.

4.5 The Reversal

In 2024–2025, the pendulum began swinging back. Harvard, Yale, Brown, Dartmouth, and Caltech reinstated mandatory SAT/ACT testing. Stanford, Cornell, and UT Austin announced testing requirements for 2025–26.

At schools that restored test requirements, applications declined, mechanically raising acceptance rates — the reverse of the test-optional inflation effect.

Among top LACs, Williams, Amherst, and Bowdoin remain test-optional as of the 2026 admissions cycle. Swarthmore was “test-optional for at least one more admissions cycle” as of Class of 2028 admissions.

Source: Cosmic.nyc, “2024-2025 College Admissions: The Year That Changed Everything,” December 2025; Top Tier Admissions, “Little Ivies: Class of 2028 Acceptance Rates,” June 2024.

4.6 Correction to Section 2 of the Original

The acceptance rates and SAT ranges in Section 2 should carry a caveat: they were generated under test-optional conditions that systematically inflated SAT ranges (by excluding low submitters) and deflated acceptance rates (by inflating application volume). Both distortions make the schools appear more selective than they would under test-required policies. This is not fraud — but it is a methodological artifact that the reader should be aware of.

Gap Fix 5: The Consortium Model — A Structural Feature, Not a Footnote

5.1 The Gap

The original document mentions Pomona’s access to Harvey Mudd once but does not analyze the consortium model as a structural feature of elite LACs.

5.2 The Three Major Consortia

The Claremont Colleges (5Cs / 7Cs): - Members: Pomona, Claremont McKenna, Harvey Mudd, Pitzer, Scripps (undergrad); Claremont Graduate University, Keck Graduate Institute (graduate) - Combined: 9,000+ students, 3,600+ faculty/staff, 2,700+ courses - All five undergraduate campuses are contiguous — 20 minutes to walk end-to-end - Students can cross-register at any college for no extra cost - A typical Pitzer student takes “half their classes at Pitzer, 40% at the other 5Cs, and 10% in the Department of Natural Sciences” - Students can declare an off-campus major at another consortium school - Shared dining halls, athletic teams, libraries, student organizations, and residential options - **2025 marks 100 years** of the consortium model

Source: Pitzer College, “The Claremont Colleges Advantage.” <https://www.pitzer.edu/academics/claremont-colleges-advantage>; Expert Admissions, “College Consortiums.” <https://expertadmissions.com/college-consortiums/>

Five College Consortium: - Members: Amherst, Hampshire, Mount Holyoke, Smith, University of Massachusetts Amherst - Combined: ~30,000 students, 2,100+ faculty, 7,000+ courses - Free bus service links all five campuses - Cross-registration, certificate programs (Asian/Pacific/American Studies, Coastal and Marine Science, Cognitive Neuroscience, International Relations, etc.) - Amherst students can access UMass Amherst’s R1 research infrastructure - Hampshire’s closure (Section 10.1) removes one member — the implications are not addressed in the original

Source: Top Tier Admissions, “Consortium Colleges: The Best of Both Worlds,” March 2023.

Tri-College Consortium (+ Quaker Consortium): - Tri-College: Bryn Mawr, Haverford, Swarthmore — three elite LACs in suburban Philadelphia - Quaker Consortium extends to include the University of Pennsylvania

(Ivy League R1) - Shuttle service between Tri-College campuses; students at LACs can access Penn’s resources
 - ~4,500 students across the three LACs

Source: IvyWise, “Exploring the Advantages of College Consortiums,” April 2025.

5.3 Why This Matters

The consortium model directly addresses Section 11 of the original (structural vulnerability — no engineering). A Pomona student has access to Harvey Mudd engineering. A Swarthmore student can take courses at Penn. An Amherst student can access UMass’s R1 labs.

This also changes the “small school” calculation: a student at Pomona is not limited to 1,700 peers and ~400 course offerings. They have access to 9,000 peers and 2,700 courses across five colleges. The social and academic world is fundamentally larger than the enrollment numbers suggest.

The consortium model is arguably the most important structural innovation in American liberal arts education — and the original document treats it as an afterthought.

Gap Fix 6: International Students — An Invisible 10%+

6.1 The Data

IPEDS data (2024–2025) for the top five LACs:

Institution	International Students	% of Total
Williams College	224	10.4%
Amherst College	~12% (per CDS)	~12%
Swarthmore College	~12% (per CDS)	~12%
Pomona College	~11% (per CDS)	~11%
Bowdoin College	~7% (per CDS)	~7%

Williams’ international students come from 60+ countries, with the largest contingents from China (32), Canada (17), Pakistan (13), United Kingdom (11), South Korea (10), India (9), and several African nations (Rwanda 7, Kenya 6, Ghana 8).

Source: Williams College Enrollment & Graduation Data, Fall 2024 Catalog; CollegeTuitionCompare.com (IPEDS data); institutional Common Data Sets.

6.2 Why This Matters

1. **Diversity statistics:** International students are categorized as “Nonresident/International” in IPEDS, separate from racial/ethnic categories. A school reporting 12% Asian students and 10% international students may have a significant overlap (or not). The SFFA analysis (Section 3E) tracks domestic racial diversity but doesn’t account for how international student composition affects the campus experience.
2. **Earnings data:** College Scorecard tracks only students who received federal financial aid — international students on institutional aid are likely excluded. This means the Scorecard earnings data systematically underrepresents the full student body’s outcomes.
3. **Post-graduation trajectories:** International students face visa constraints (OPT, H-1B lottery) that channel them into specific industries (tech, finance, consulting) or back to home countries. First-destination

data (Section 7.1) that doesn't disaggregate international students may conflate visa-driven career choices with genuine preference.

4. **Financial model:** Many LACs are need-aware for international applicants even when need-blind for domestic students. International full-pay students subsidize the aid budget. This creates a tension that is invisible in the document.
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Gap Fix 7: Study Abroad — A Distinctive Feature, Absent

7.1 The Data

Study abroad participation at top LACs runs significantly above the national average (~16% for all bachelor's students, per IIE Open Doors data):

- **Swarthmore:** "More than half of the students at Swarthmore College take part in the study abroad program" — programs in 300+ locations
- **Swarthmore (alumni account):** "Roughly 40% of students study abroad... it's extremely common"
- **Williams:** ~50% study abroad (institutional reports)
- **National LAC average:** Higher than R1 universities, though exact sector-level data varies

Study abroad is relevant because: (a) it's a distinctive educational experience that large universities deliver at lower per-capita rates due to logistics; (b) it temporarily reduces campus population, creating a different junior-year dynamic; (c) financial aid travels — at need-blind LACs, students pay Swarthmore tuition and Swarthmore covers the abroad program costs.

Sources: Swarthmore College, Admissions Consultants profile; Reddit r/Swarthmore (alumni account, 2024); IIE Open Doors Report.

Gap Fix 8: The NSSE Gap — Engagement During College

8.1 The Gap

The original uses Gallup alumni surveys (retrospective, years after graduation) but omits NSSE, which measures engagement *during* college.

8.2 What NSSE Is

The National Survey of Student Engagement (NSSE) is administered by Indiana University's Center for Postsecondary Research. Since 2000, nearly 1,700 institutions have participated. It surveys first-year and senior students on:

- **10 Engagement Indicators** across four themes: Academic Challenge, Learning with Peers, Experiences with Faculty, Campus Environment
- **6 High-Impact Practices (HIPs):** Service-learning, learning communities, research with faculty, internships, study abroad, culminating senior experience

NSSE has shown:

“Increases in Engagement Indicators — especially Effective Teaching Practices, Quality of Interactions, and Supportive Environment — are associated with an increased likelihood of first-year retention.”

“Engagement has a positive relationship with grades and persistence, and more pronounced effects for historically underserved and underrepresented students.”

Source: NSSE Essentials, Indiana University. <https://nsse.indiana.edu/research/nsse-essentials/index.html>

8.3 LACs and NSSE

NSSE uses “National Liberal Arts Colleges” as a comparison group (77 institutions as of 2024). While individual institutional NSSE results are proprietary (schools choose whether to publish), the structural characteristics of LACs — small classes, high faculty interaction, required research, senior capstone projects — align directly with the HIPs that NSSE measures.

The Wabash National Study of Liberal Arts Education, which triangulated NSSE data with other records, demonstrated that engagement indicators “predict gains in critical thinking, literacy, personal well-being, and other outcomes.”

Critical caveat: NSSE is descriptive, not outcomes-based. It measures what students *do* and *perceive*, not what they *learn*. And individual institutional results are not publicly comparable unless schools opt in.

Source: Wabash College, “The National Survey of Student Engagement (NSSE)”;
NSSE, “About NSSE.” <https://nsse.indiana.edu/nsse/about-nsse/index.html>

8.4 Why This Matters

The Gallup finding in Section 12 — that institutional type “hardly matters at all” for workplace engagement — is based on retrospective alumni self-report. NSSE data collected *during* college consistently shows that the educational practices LACs emphasize (faculty interaction, undergraduate research, collaborative learning) are the same practices associated with better engagement and retention. The gap between “LACs have the structural conditions for the Big Six” and “Gallup finds no institutional-type effect” may be partly explained by NSSE’s finding that participation in HIPs, while higher at LACs, is still not universal — roughly 3 in 5 first-year students and seniors had participated in at least two HIPs as of 2019.

Gap Fix 9: The Pre-Professional Paradox — What “Liberal Arts” Actually Means Now

9.1 The Gap

The original document presents these facts without connecting them: CS is the #1 major at Pomona (13%), 26.4% of Williams graduates enter finance, and career funneling is structurally manufactured.

9.2 The Tension

The core identity claim of a liberal arts college is breadth: students study across disciplines, develop transferable skills, and defer specialization. But the empirical reality at top LACs in 2026:

- **Pomona’s top 5 majors:** Computer Science (13%), Economics (10%), Mathematics (7%), Neuroscience (7%), Molecular Biology (5%) — four of five are STEM
- **Williams Class of 2023:** 26.4% entered finance, the single largest destination sector
- **Harvard Class of 2025:** 53% entered finance (21%), tech (18%), or consulting (14%)
- **Swarthmore:** Engineering is the most popular intended major for the Class of 2028, followed by CS, political science, English, and biology

The distribution requirements remain. Williams students must take courses across three divisions. Pomona requires breadth. But the *experienced* trajectory is increasingly: declare a STEM major, optimize for quant skills, recruit into finance or tech.

9.3 The Honest Assessment

Two interpretations are defensible:

Interpretation A (charitable): The liberal arts model works *because* it produces CS majors who also studied philosophy, art history, and political theory. The breadth requirement ensures these graduates are different from CS majors at Georgia Tech or Purdue — and the Revelio Labs finding that LAC graduates earn a premium within the same major supports this.

Interpretation B (skeptical): The “liberal arts” label has become a brand wrapper around what is functionally a small, expensive pre-professional institution. The distribution requirements provide a veneer of breadth but the 4-year arc — internship optimization, finance recruiting, career funneling — is structurally identical to what happens at Penn or Duke. The value proposition is not “liberal arts” but “small, wealthy, highly selective, well-connected private school.”

Neither interpretation is fully correct. But the document should name the tension rather than leaving the reader to discover that the #1 major at the #5 liberal arts college is Computer Science.

Gap Fix 10: Dale-Krueger’s Temporal Validity — Does the Null Result Still Hold?

10.1 The Gap

The original cites Dale-Krueger (2002, 2014) as the central methodological check on selectivity effects but doesn’t note that their data comes from students who enrolled in 1976 and 1989.

10.2 The Hoxby Critique

Caroline Hoxby (Stanford) offered the sharpest methodological critique:

“Dale and Krueger (2002) compute lower rates of return but their estimates are based on an identification strategy that is much less credible. They compare students who gained admission to approximately the same

menu of colleges... Since at least 90 percent of students who have the same menu similarly choose the more-selective college(s) within it, the strategy generates estimates that rely entirely on the small share of students who make what is a very odd choice.”

Hoxby’s argument: the students who *chose* the less selective school from a comparable menu are systematically unusual — they are characterized by omitted variables that affect both their college decision and their outcomes. The null result may reflect the oddity of those students, not the irrelevance of selectivity.

Source: Hoxby CM, “The Changing Selectivity of American Colleges” (survey article); discussed in EconLib, “Hoxby vs. Dale-Krueger on the Selectivity Premium,” November 2013.

10.3 The Temporal Problem

The labor market has changed fundamentally since the 1976 and 1989 cohorts Dale-Krueger studied:

1. **Prestige screening has intensified.** Rivera’s *Pedigree* (2015) documents how elite firms now use institutional prestige as the *first* filter — “the first cut in EPS interviewing is effectively completed four years in advance by school admissions committees.” This mechanism was less developed in the 1970s–1990s.
2. **Network effects compound over time.** Social media, alumni databases, and LinkedIn have made institutional networks more accessible and more valuable than when the Dale-Krueger cohort graduated.
3. **The MBA/grad school pipeline:** Many elite LAC graduates use their undergraduate credential to access elite MBA and law programs. If the selectivity premium operates primarily through graduate school placement (which the original document’s data on graduate degree attainment supports), Dale-Krueger’s earnings-only approach may miss the most important channel.
4. **The counterpoint:** Inside Higher Ed’s analysis (2019) anticipated this concern: “The nature of the student bodies at selective colleges and the degree of selectivity has changed dramatically since that time. Many students who 20 years ago would have been admitted to multiple highly selective colleges without question can’t sniff a spot in today’s hyperselective admissions landscape. Will those changes weaken or strengthen the Krueger-Dale conclusions?”

No one has replicated Dale-Krueger with post-2000 cohort data. The Chetty studies (2020, 2023) use more recent data and find *causal* effects of elite college attendance — but use a different methodology (internal admissions data, not the Dale-Krueger “matched menu” design). This gap in the literature means the original document’s reliance on Dale-Krueger as the definitive word on selectivity is overweighted.

Sources: Inside Higher Ed, “Alan Krueger Made an Important Contribution,” April 2019; EconLib, “Hoxby vs. Dale-Krueger,” November 2013; Chetty R, Deming DJ, Friedman JN, “Diversifying Society’s Leaders?” *QJE*, 2023.

Gap Fix 11: Geographic Concentration and Its Implications

11.1 The Pattern

Of the top 10 LACs in the US News 2026 rankings, the geographic distribution is:

Region	Schools
Rural/suburban New England	Williams (MA), Amherst (MA), Bowdoin (ME), Wellesley (MA), Middlebury (VT)
Suburban Mid-Atlantic	Swarthmore (PA), Haverford (PA)
Suburban Southern California	Pomona (CA), Claremont McKenna (CA)

All are in the Northeast or Southern California. Zero are in the South, Midwest, or Mountain West (Grinnell and Carleton are notable exceptions outside the top 5–7).

11.2 Why This Matters

1. **Internship access:** Students at Williams (Williamstown, MA, population ~7,000) face a 3-hour drive to Boston and a 3.5-hour drive to NYC. Pomona students are 35 miles from downtown LA. This creates an uneven playing field for summer and term-time internships — a structural disadvantage for rural New England LACs that the career funneling mechanism (Section 7.3) partially compensates for through on-campus recruiting.
2. **Cultural accessibility:** For students from the South, Midwest, or West, a small New England town can feel culturally alien. The regional concentration of top LACs limits the geographic diversity of applicant pools and contributes to the overrepresentation of Northeastern families.
3. **Weather and isolation:** Brunswick, Maine (Bowdoin) and Williamstown, MA (Williams) are genuinely isolated in winter. This is a feature for some students and a dealbreaker for others — but it’s a real factor in the college experience that “rankings” don’t capture.

Gap Fix 12: Credential vs. Education — The Unspoken Distinction

12.1 The Gap

The original document oscillates between evaluating LACs as *educational experiences* (Gallup, mentorship, Big Six, undergraduate research) and as *credentials/sorting mechanisms* (career funneling, Rivera, prestige hiring, Goldman Sachs placement). These are fundamentally different value propositions and the document should name the distinction.

12.2 The Two Value Propositions

The education proposition: “At a LAC, you will learn to think critically, write clearly, reason across disciplines, and develop deep mentoring relationships with faculty. These skills will serve you throughout your career, regardless of your first job.”

This is what the Gallup “Big Six” evidence supports. It’s what the PhD production data reflects. It’s the theoretical argument for liberal arts in the age of AI. And it’s what the \$135,600 per-student expenditure at Williams is ostensibly buying.

The credential proposition: “A degree from Williams/Amherst/Pomona signals to employers, graduate schools, and professional networks that you survived a sub-10% admissions filter. This credential opens doors — finance recruiting, T14 law school, elite MBA programs — that would be harder to access from other institutions.”

This is what Rivera’s *Pedigree* documents. It’s what the career funneling research shows. And it’s what the Chetty data suggests causally matters (especially for URM students).

12.3 Why the Distinction Matters

If the value is primarily educational, then Dale-Krueger is correct — a motivated student can get a comparable education at many institutions, and the specific institution matters less than the student’s engagement with it. If the value is primarily credentialing, then Dale-Krueger is potentially wrong for the current era — because the credential premium has intensified as prestige screening has become more systematic (Rivera, 2015).

Most students and families are buying *both* — but the ratio matters for the decision. A student whose goal is to become a philosophy professor should optimize for education (faculty mentorship, research opportunities, PhD placement). A student whose goal is Goldman Sachs should optimize for credential (on-campus recruiting, alumni network, brand recognition).

The honest answer is that top LACs deliver both, but their marketing emphasizes the education proposition while their outcomes data increasingly reflects the credential proposition.

Gap Fix 13: Chetty Mobility Data — Quantifying the SFFA Impact

13.1 The Gap

Section 3E.4 of the original qualitatively argues that SFFA removes the population that benefits most from elite attendance. Here is a more precise framing.

13.2 The Chetty Framework

Chetty’s Mobility Report Cards (NBER Working Paper 23618, published in *QJE* 2020) define:

- **Access rate:** % of students from the bottom income quintile
- **Success rate:** % of bottom-quintile students who reach the top quintile
- **Mobility rate:** Access rate $\times$ Success rate

Key findings for elite private colleges (“Ivy-Plus”):

“Children whose parents are in the top 1% of the income distribution are 77 times more likely to attend an Ivy League college than those whose parents are in the bottom income quintile.”

“Children from low and high-income families have similar earnings outcomes conditional on the college they attend, indicating that low-income students are not mismatched at selective colleges.”

“Mid-tier public universities such as the City University of New York and California State colleges tend to have the highest rates of bottom-to-top quintile mobility” — because they have far higher access rates despite lower success rates.

“Rates of upper-tail (bottom quintile to top 1%) mobility are highest at elite colleges” — they produce the biggest leaps for the few low-income students who attend.

13.3 The Quantification

At “Ivy-Plus” institutions, approximately **4.7% of students** come from the bottom income quintile (Chetty data). Among those, the success rate (reaching top quintile) is high — roughly 60% at schools like Columbia. Post-SFFA, the Brookings data shows a **16.3% decline in Black freshmen** at highly selective institutions. Since Black students at these institutions disproportionately come from lower-income backgrounds, SFFA is reducing the access rate for precisely the subgroup with the highest marginal benefit from attendance.

Order-of-magnitude estimate: If elite LACs collectively enrolled ~500 Black freshmen per year pre-SFFA, and enrollment dropped 16.3%, that’s ~80 fewer Black freshmen per year across the sector. If the Chetty success rate for bottom-quintile students at elite institutions is ~50–60%, and a meaningful fraction of those 80 students would have been bottom-quintile, the SFFA ruling is preventing roughly 20–40 students per year from making the bottom-to-top quintile leap that elite college attendance would have enabled.

This is a small number in absolute terms — but these are individual lives where the causal effect of institutional attendance is strongest.

13.4 The Counterargument

Chetty also found that mid-tier public universities have the *highest overall* mobility rates because their access rates are so much higher. A Black student who attends CUNY instead of Amherst still has a meaningful shot at upward mobility — just not as dramatic a leap. The SFFA cascade effect (Section 3E.3 of the original) may be redistributing URM students to state flagships where mobility rates are lower per-student but higher in aggregate.

Sources: Chetty R et al., “Mobility Report Cards: The Role of Colleges in Intergenerational Mobility,” NBER Working Paper 23618 (2017), *QJE* 135(3) (2020); Brookings Institution (April 2026), cited in original Section 3E.3.

Gap Fix 14: Evidence Quality Downgrades

14.1 Section 5: Business Leadership — Recommend Downgrade

The Grinnell study (Section 5.1) is an opinion essay by a college communications employee published on Inside Higher Ed in 2012. It is not peer-reviewed. The Fortune 50 CEO sample (N=54) is statistically meaningless. The S&P 500 CEO data is from 2004, reflecting graduates of the 1960s–1980s.

Recommended label: “Historical/Anecdotal — No rigorous evidence exists for the LAC-to-leadership pipeline claim.”

14.2 Section 3D.1: Medical School Rates — Recommend Stronger Caveat

The “qualified applicant” filter at Williams ($\text{GPA} \geq 3.7$, $\text{MCAT} \geq 512$) is above the 85th percentile of all medical school applicants. Any pool pre-filtered to that level would have high acceptance rates from a community college. These self-reported, non-standardized rates are institutional marketing, not comparable data.

Recommended label: “Self-reported institutional marketing — not comparable across institutions or to national average without controlling for applicant qualification.”

14.3 Section 7.2: Swarthmore Alumni Survey — Recommend Vintage Warning

The 86% graduate degree attainment for the Class of 1969 reflects 1970s America. The PhD premium, academic job market, and cost of graduate education have changed fundamentally. This number should be presented as historical context, not current evidence.

14.4 Section 2: Rankings Table — Recommend Transparency Fix

The table lists ranks 1, 2, 4, 5, 7 and labels them “Top 5.” The reader doesn’t know what’s at ranks 3 and 6. The text says “stripping out military academies” — the table should explicitly note: “Ranks 3 and 6 (U.S. Naval Academy and U.S. Military Academy, respectively) excluded as military service academies.” (Or whichever institutions actually hold those ranks — verify against the actual 2026 USNWR list.)

Source Index (Additions to Original)

ID	Source	Type	Date
S43	Bloomberg, “Top US College Costs Squeeze Middle Class”	Data journalism	April 2025
S44	Neumuller S, “The Real Cost of a Top-tier Liberal Arts Education,” <i>Mildly Efficient</i>	Economic analysis	September 2024
S45	Bowdoin College, “Net Price by Income”	Institutional data	Current
S46	Healthy Minds Study 2024–2025	Survey of 84,735 students	2025
S47	<i>Bowdoin Orient</i> , “Banded together” (NESCAC athletics investigation)	Student journalism	2014
S48	Williams College, Fast Facts	Institutional data	Fall 2025
S49	NESCAC, Directors’ Cup Final Standings	Administrative data	June 2025
S50	Hsu G, Sharkey A, <i>American Sociological Review</i> (test-optional)	Peer-reviewed	2025
S51	Hoxby CM, critique of Dale-Krueger	Survey article	2009
S52	NSSE, Indiana University	Survey instrument	Ongoing since 2000
S53	Pitzer/Claremont Colleges consortium data	Institutional	Current
S54	Top Tier Admissions, “Consortium Colleges”	Journalism	March 2023
S55	Wallace JB, <i>Never Enough</i>	Published book	2023
S56	NPR, “Colleges targeting financial aid to middle-class families”	Journalism	October 2024
S57	Chetty R et al., “Mobility Report Cards,” NBER WP 23618	Peer-reviewed	2017/2020
S58	Oriel Admissions, Williams acceptance rate analysis	Admissions consulting	May 2026
S59	EdWorkingPapers, test-optional study	Working paper	2024
S60	CollegeTuitionCompare / IPEDS, Williams enrollment	Federal data	2024–25

These notes supplement the original research document. All quoted passages are verbatim from cited sources. Interpretive claims are flagged as such.